Exam Date & Time: 25-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH (AUTOMOBILE ENGINEERING) END SEMESTER EXAMINATIONS, NOV 2024

STRENGTH OF MATERIALS [AAE 2126]

Marks: 50 **Duration: 180 mins.**

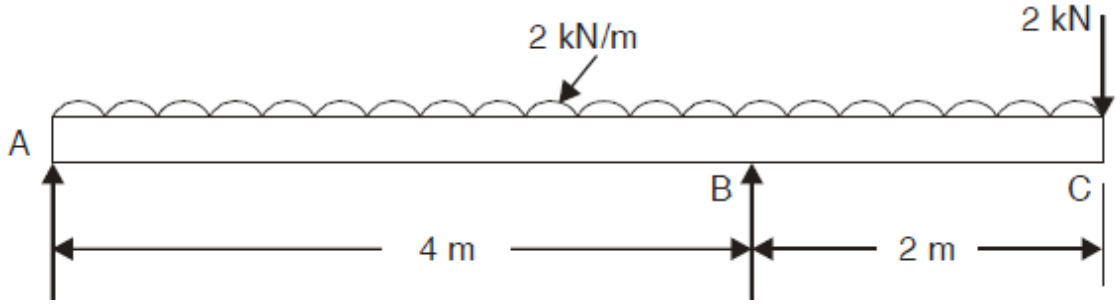
A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) Draw the shear force and bending moment diagrams for the overhanging beam carrying a uniformly distributed load of 2 kN/m over the entire length and a point load of 2 kN as shown in Figure 1. Locate the point of contraflexure.

A)



(5)

Figure 1.

B) A simply supported beam 6 m long is carrying a uniformly distributed load of 5 kN/m over a length of 3 m from the right end. Draw shear force and bending moment diagrams for the beam. (3)

C) A brass strip, 80 mm wide and 30 mm thick, is bent into the arc of radius 60 m. Determine the maximum stress developed in the strip? Take E = 1 X 10^5 N/mm^2. (2)

2) A thin cylindrical shell is 100 cm in diameter, 1 cm thick and 5 m long with internal pressure of 3 N/mm^2. Determine the change in diameter, change in length and change in volume. Take E = 2X10^5 N/mm^2 and μ =0.3. (5)

A)

B) The dimension of the T section is shown in Figure 2. A cantilever beam of 4m long is of T-section. Determine point load w at the free end so that the maximum stress in the section does not exceed 90 MPa. (3)

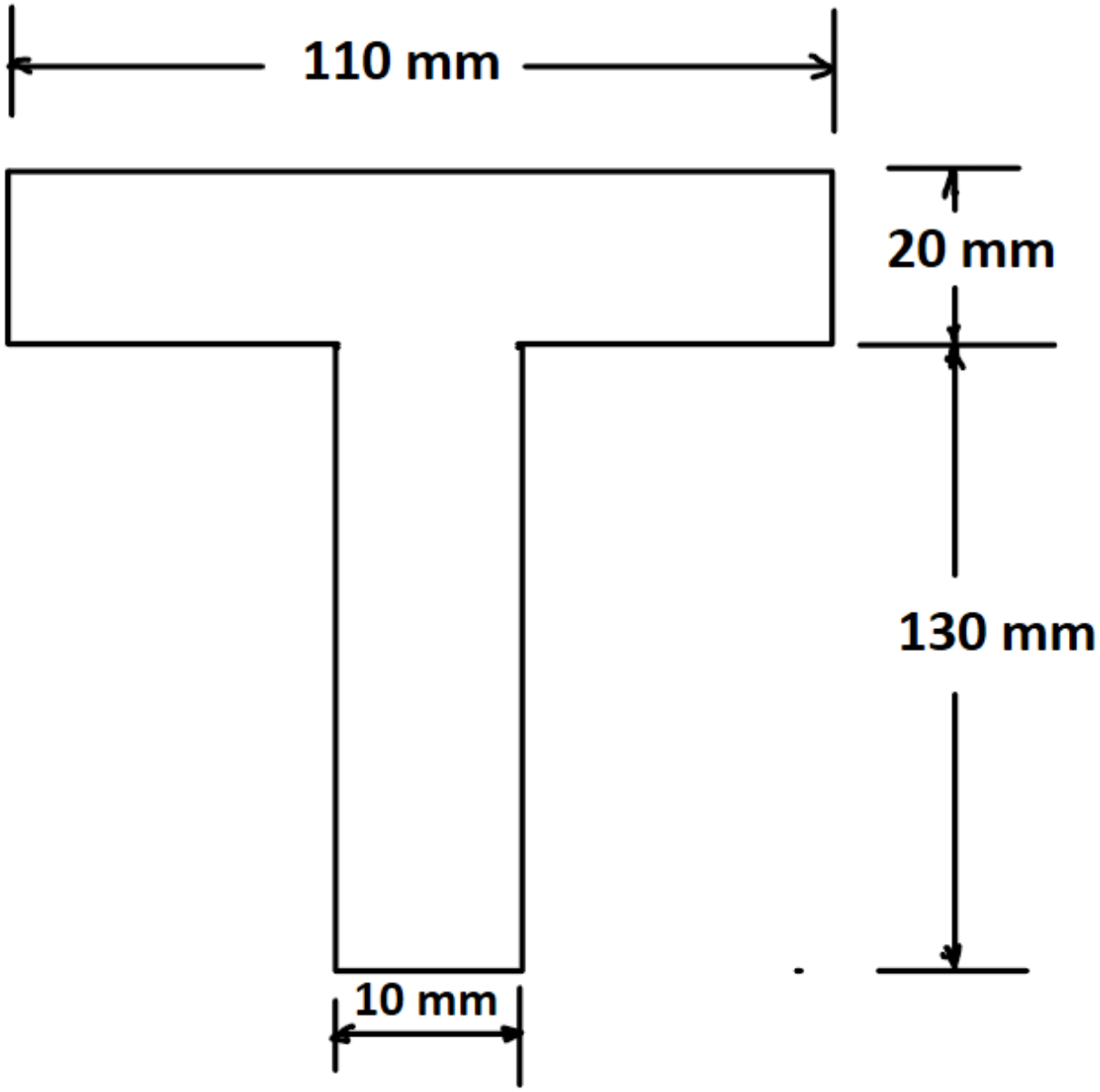


Figure 2

C) A cantilever beam is rectangular in section having 80 mm width and 120 mm depth. If the cantilever is subjected to a point load of 6 kN at the free end and the bending stress is not to exceed 40 MPa, determine the span of the cantilever beam. (2)

3) The shear force acting on a section of a beam is 50 kN. The section of the beam is T-shaped with dimensions 100 mm x 100 mm x 20 mm as shown in Figure 3. The moment of inertia about the horizontal neutral axis is 314.221 x 10^4 mm^4. Calculate the shear stress at the neutral axis and the junction of the web and the flange. (4)

A)

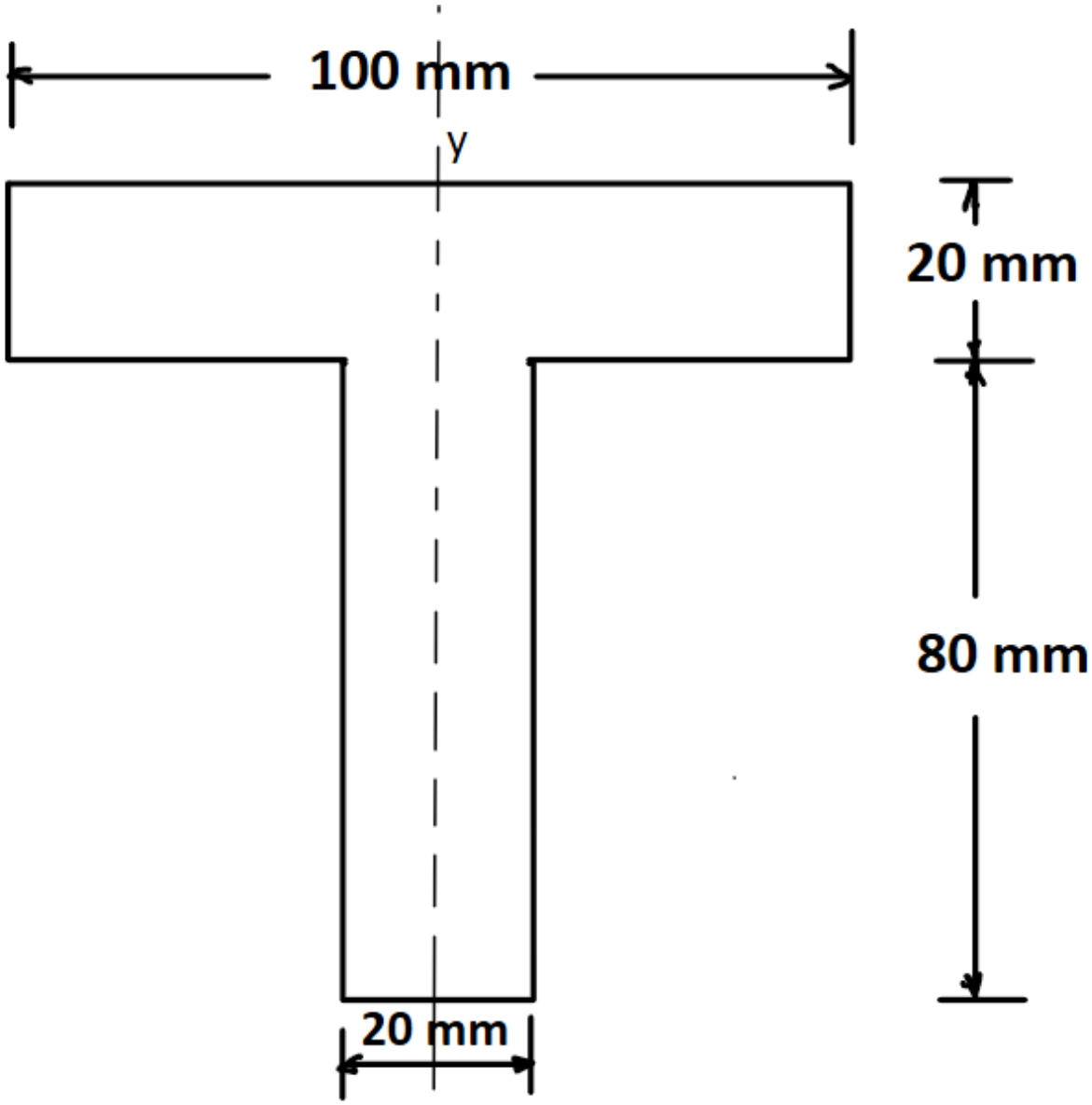


Figure 3

B) Prove that the maximum shear stress in a circular section of a beam is 4/3 times the average shear stress. (3)

C) A solid steel shaft transmits a torque of 10 kN-m. If the shearing stress is not to exceed 45 MPa, find the minimum diameter of the shaft. (3)

4) A beam of length 6 m is simply supported at its ends and carries two point loads of 48 kN and 40 kN at a distance of 1 m and 3 m respectively from the left support. By using Macaulay's method find deflection under each load. Given $E = 2 \times 10^5$ N/mm^2 and $I = 85 \times 10^6$ mm^4 (4)

A)

B) A shaft of 400 mm in length and 50 mm in diameter is bored for a part of its length to a 20 mm diameter. Find the maximum power the shaft can transmit at a speed of 180 rpm (4)

with a limiting shear stress of 80 N/mm^2. Also, determine the length of the shaft that has to bear a bore of 20 mm diameter if the angle of twist is uniform in the entire length of the shaft.

C) In a hollow circular shaft of outer and inner diameters of 20 cm and 10 cm respectively, the shear stress is not to exceed 40 N/mm^2. Find the maximum torque that the shaft can safely transmit. (2)

5) A pipe of 200 mm internal diameter and 50 mm thickness carries a fluid at a pressure of 10 N/mm^2. Calculate the maximum and minimum intensities of circumferential stresses across the section. Also, sketch the radial stress (pressure) distribution and circumferential stress distribution across the section. (4)

A)

B) A cantilever beam 120 mm wide and 150 mm deep is 1.8 m long. Determine the slope and deflection at the free end of the beam, when it carries a point load of 20 kN at its free end. Take E for the cantilever beam as 200 GPa. (4)

C) A vessel in the shape of a thin spherical shell of 1.2 m internal diameter and 12 mm shell thickness is subjected to pressure of 1.6 N/mm^2. Determine the stress induced in the material of the vessel. (2)

-----End-----